

Skill & Product Library / Sales Automation

AI Agent System

Live in production

AI Customer Agent Suite

A production website AI agent, plus the qualification pipeline, reporting warehouse, and revenue-attribution layer that prove exactly what it's worth.



Quarter-million-dollar deal sourced from a single chat — traced to the exact conversation

HubSpot

Breeze Customer Agent

n8n

Azure OpenAI

GPT-4o-mini

Microsoft Graph

SharePoint

MQL classification

revenue attribution

LangChain

Overview

A production AI customer agent for a global HR-tech (EOR) platform (~500 employees, hiring in 160+ countries) — built end to end by one person: the agent itself (versioned conversation guidelines, guardrails, knowledge base, routing), a live AI qualification pipeline that classifies every closed chat as MQL or Not under strict governance rules, a three-workflow reporting warehouse, and a three-tier deal-attribution layer. Live on the pricing page since launch. Headline result: a visitor chatted one day, the pipeline stamped an MQL, and a rep opened a **quarter-million-dollar deal the next day** — traceable to the exact conversation. **Over \$500K in agent-sourced deal value** in the first four months.

The Problem

The company needed an AI agent on its website that qualifies visitors, feeds the CRM cleanly, and — the hard part — can *prove* its revenue impact. Chatbot leads were unattributed, duplicated, and contaminated by hundreds of legacy pre-launch contacts, so nobody could answer "what is the chatbot worth?"

How It Works

1. **HubSpot Customer Agent (Breeze)** — HubSpot's native AI chat product — runs the agent on the pricing page under versioned conversation guidelines governing what it says and when it routes to sales, partners, or support.
2. On every closed conversation, a webhook fires an **n8n** qualification workflow (4,146 logged executions) — n8n is a visual workflow-automation platform that orchestrates the whole pipeline.
3. A hand-written **JavaScript normalizer** strips HTML, orders messages, distinguishes visitor from bot by actor ID, and extracts email/phone/company from free text with personal-domain filtering — deterministic cleanup before any AI sees the data.
4. An **Azure OpenAI GPT-4o-mini** agent (Microsoft-hosted OpenAI model) with a structured-output parser classifies MQL vs Non-MQL under authored rules: only visitor messages count as intent; confidence < 0.65 forces Non-MQL; blank values never overwrite existing CRM data. MQLs get lifecycle + 15 chatbot properties stamped via the **HubSpot CRM APIs**, plus a drafted follow-up email.
5. A **three-workflow reporting warehouse** — webhook event collector, daily reconciliation job, and hourly AI-credit watcher (4-tier thresholds: 60/75/85/90%) — writes to a five-table SharePoint Excel workbook via **Microsoft Graph**, Microsoft's API for Office 365 files and mail.
6. A **deal-attribution walk** (deal → contact → chatbot properties) assigns three honesty tiers — sourced, influenced, excluded — operationalized as a weekly n8n stamper writing attribution tier + originating conversation onto deals, report-only until sign-off.



Stack & Integrations

Tool	Role	Access needed
HubSpot Customer Agent (Breeze)	The visitor-facing agent	Portal admin (chatflows, knowledge base)
n8n cloud	Pipeline orchestration (LangChain agent nodes)	n8n workspace; workflow import rights
Azure OpenAI (GPT-4o-mini)	MQL classification + email drafting	Azure OpenAI deployment + API credential
HubSpot Conversations + CRM v3/v4 APIs	Threads, contacts, deals, property writes	Private-app token: conversations read, contacts/deals read-write
Microsoft Graph	SharePoint Excel warehouse + Outlook alerts	OAuth2: file read-write, mail send
Python dashboard	Local reporting views	Read access to warehouse workbook

Setup Process

1. Configure the Customer Agent chatflow, knowledge base, and conversation guidelines in HubSpot.
2. Run the property-creation script to create the custom chatbot contact properties (idempotent — skips existing).
3. Build the attribution stamp workflow in HubSpot (engagement flag + source, never overwriting native source fields).
4. Import the MQL classification workflow JSON into n8n; bind HubSpot private-app and Azure OpenAI credentials; wire the shared error workflow.
5. Register the conversation-close webhook; deploy the warehouse workflows and SharePoint workbook.
6. Dry-run on a sample of closed conversations; verify stamps on live contacts before enabling writes.
7. Deploy the attribution stamper in report-only mode; enable deal writes only after sign-off.



What You Need

- HubSpot portal with Customer Agent (Breeze) enabled + private-app token
- Azure OpenAI deployment (GPT-4o-mini)
- n8n cloud workspace; Microsoft 365 (SharePoint + Outlook) OAuth credential
- Workflow JSONs + property script · **Time-to-deploy: ~2-3 weeks** including pilot

Results

- **A quarter-million-dollar open deal sourced by the agent** — visitor chat one day, rep-opened deal the next, traced to the exact originating conversation
- **Over \$500K in agent-sourced deal value** in the first four months; **~\$680K** year-to-date closed-won on chatbot-engaged contacts (influence tier)
- **~\$3.6M lifetime deal value** across ~390 chatbot-engaged contacts; 4,146 logged pipeline executions

Challenges & Fixes

- Hundreds of legacy pre-launch contacts contaminated metrics → excluded tier + launch-era cutoff; even a lost first-attributed deal stays counted (influence, not cherry-picking)
- Webhook-only reporting missed late closures → hybrid design: webhook collector *and* daily reconciliation rebuild
- Over-eager AI classification risk → confidence gate (<0.65 = Non-MQL), visitor-messages-only intent, never-overwrite rule

- "Why did this chat not create a contact?" → reconciliation-gap metric answered by transcript email search

Demonstrates

Full-stack AI product ownership (agent → pipeline → warehouse → attribution) · LLM governance rails in a production CRM · revenue-attribution design · executive-grade reporting.

STACK & INTEGRATIONS

HubSpot Customer Agent (Breeze)

HubSpot Conversations API

HubSpot CRM v3/v4 APIs

n8n cloud

Azure OpenAI

Microsoft Graph (SharePoint Excel + Outlook)

Python dashboard



WHAT IT NEEDS

- ✓ HubSpot portal with Customer Agent (Breeze) enabled
- ✓ HubSpot private-app token (conversations read; contacts/deals read-write)
- ✓ Azure OpenAI deployment (GPT-4o-mini)
- ✓ n8n cloud workspace
- ✓ Microsoft 365 OAuth credential (SharePoint file read-write, mail send)

DISCUSS THIS BUILD

Happy to walk through the architecture, the failures along the way, or how this could translate to your team.

Email Patrick

